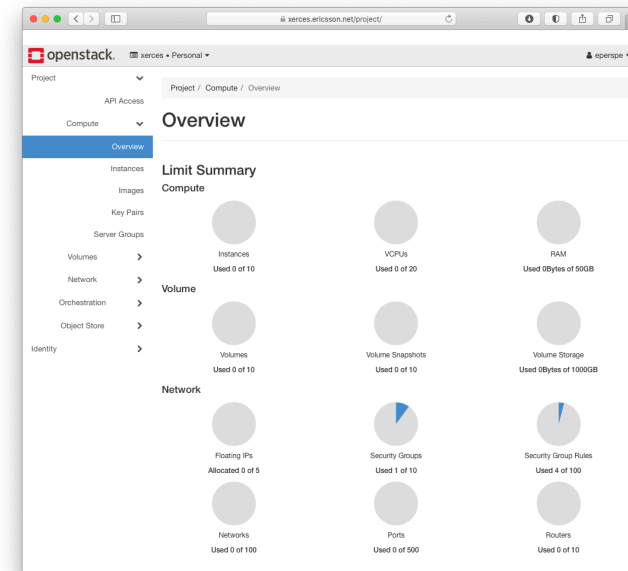
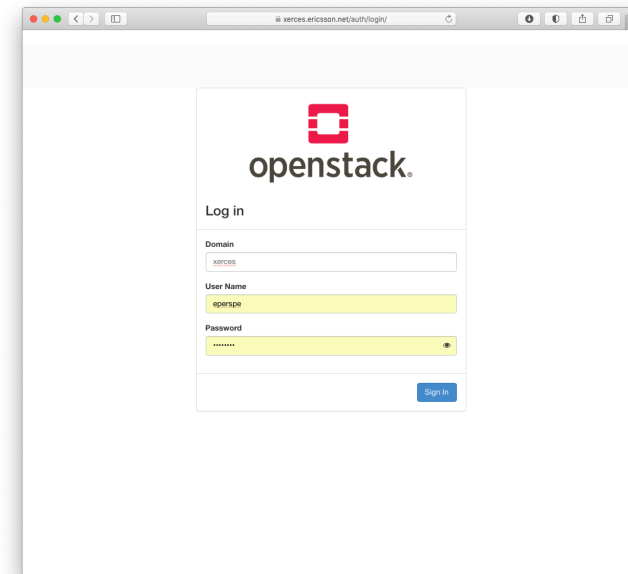


Manually setting up a cloud service.



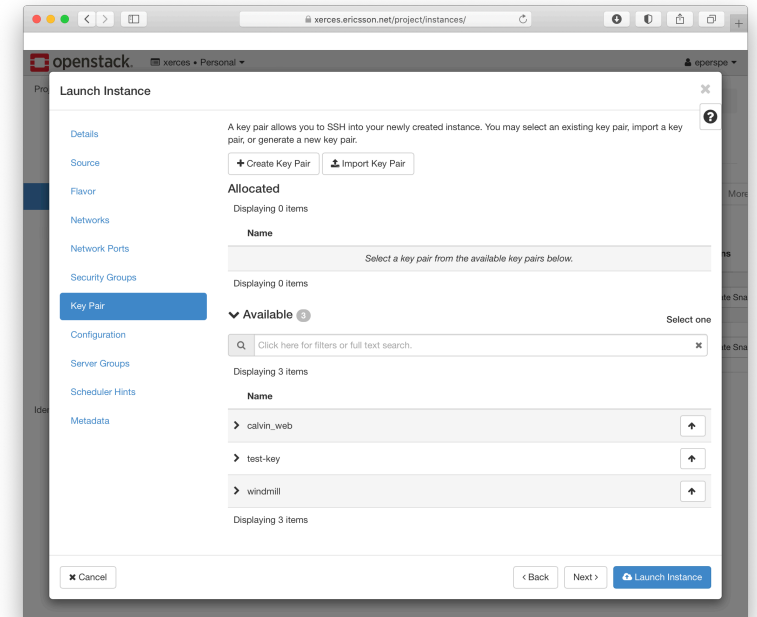
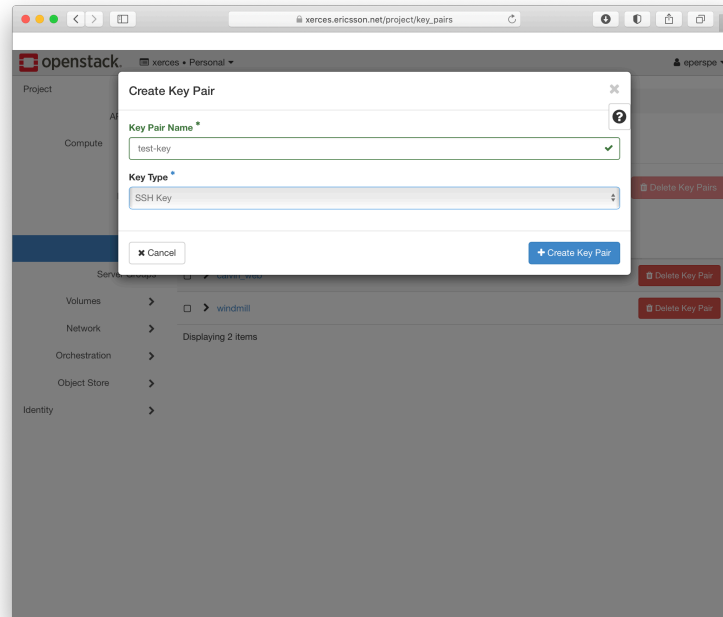
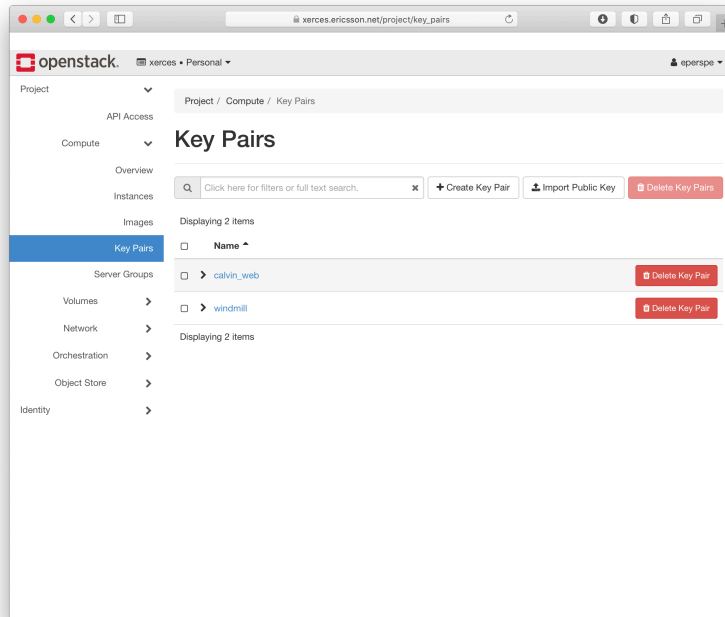
Log into Xerces

- Log into Xerces via the Horizon GUI
 - <https://xerces.ericsson.net>
 - use credentials provided for this course
- Select Compute -> Overview
 - Here you can see the available compute resources
 - Right now there is not much to see, let's fix that!



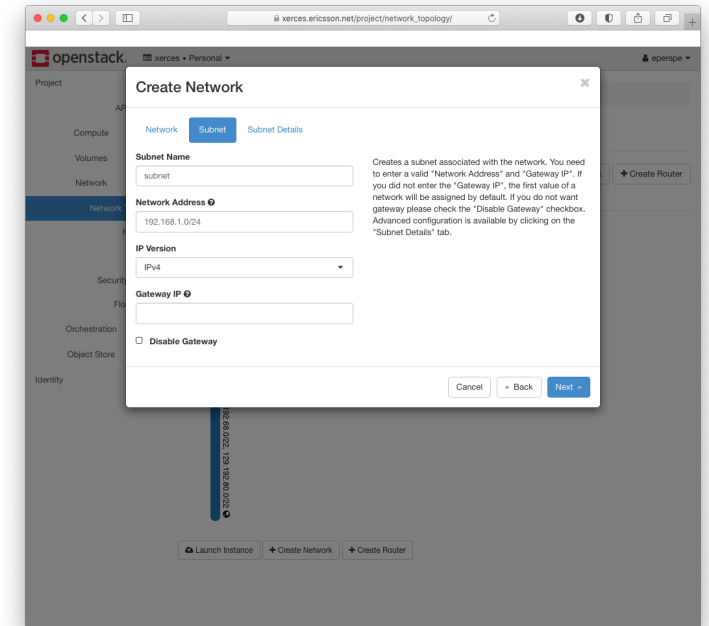
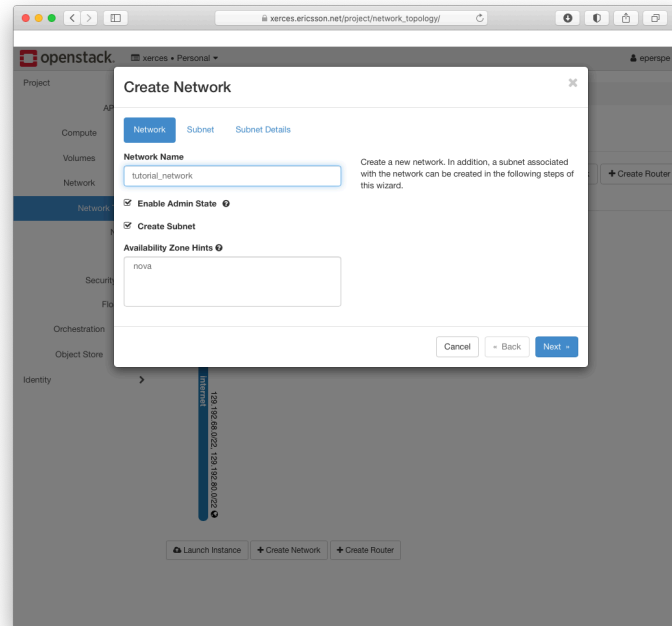
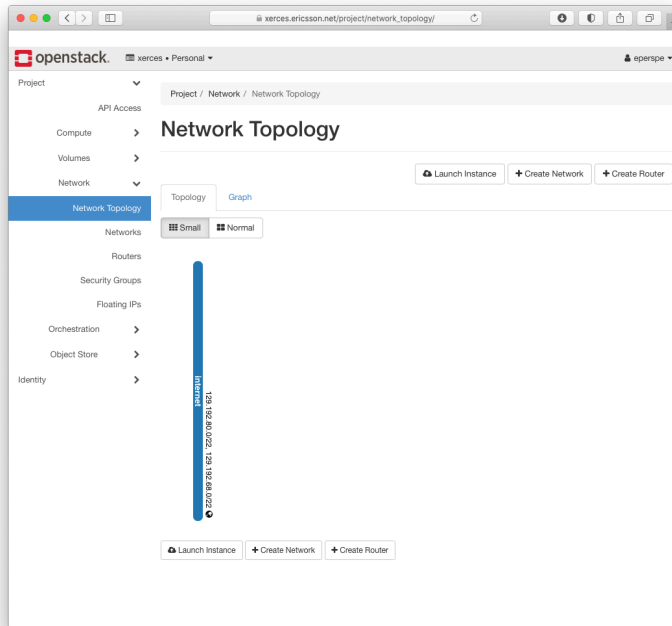
Generate a key pair

- Select Compute -> Key Pair and click "Create Key Pair"
- In the dialog enter **test-key** as the name, select **SSH Key** for Key Type, and press "Create Key Pair"
- Your private key was downloaded automatically (locate it and keep it safe)
- You can examine the public key by clicking "test-key"



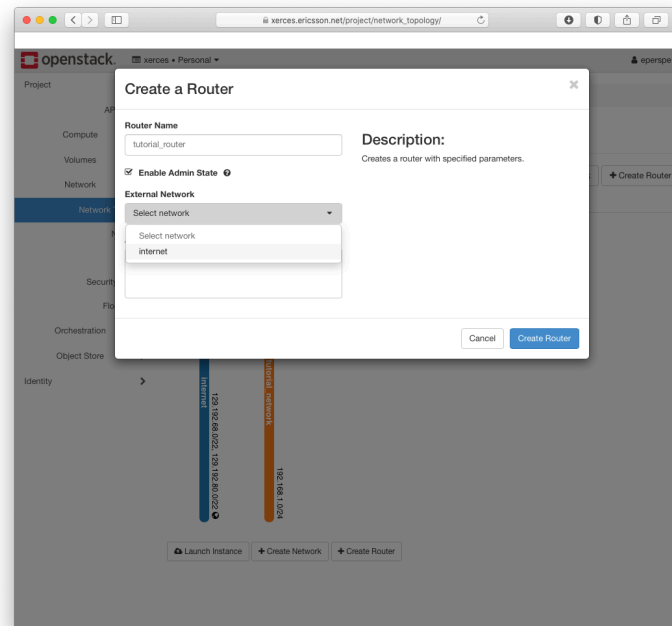
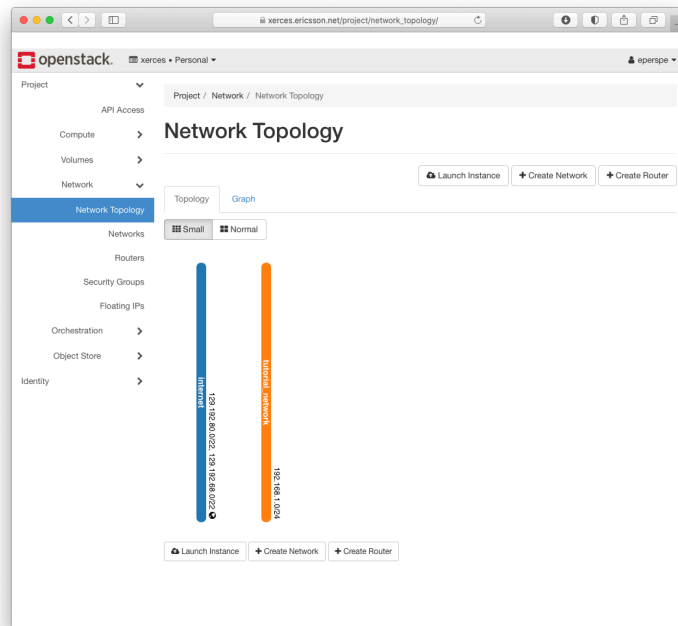
Create the networking infrastructure

- Select Network -> Network Topology. Here you can see a single network: "internet"
- Click "Create Network" to create a new network
 - Enter **tutorial_network** as the name, and click "next" to create a subnet
 - Enter subnet as the name, **192.168.1.0/24** as Network Address, and click "next"
 - Leave the subnet details unchanged, and click "create"



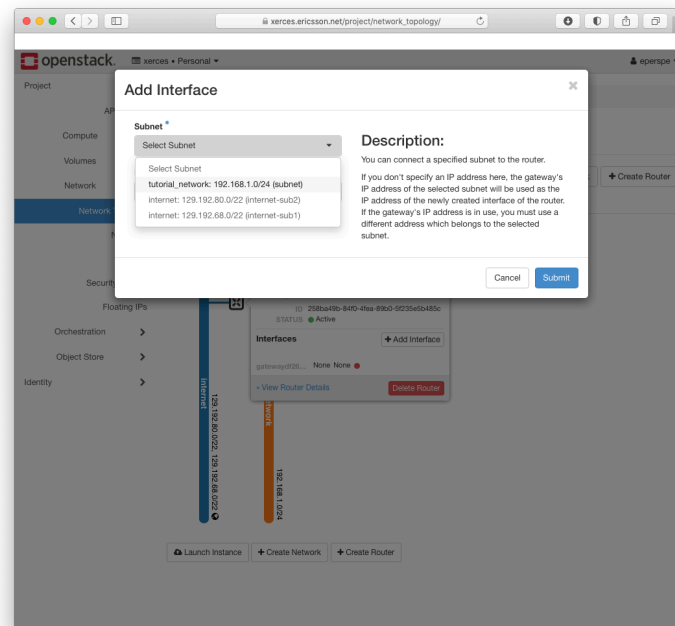
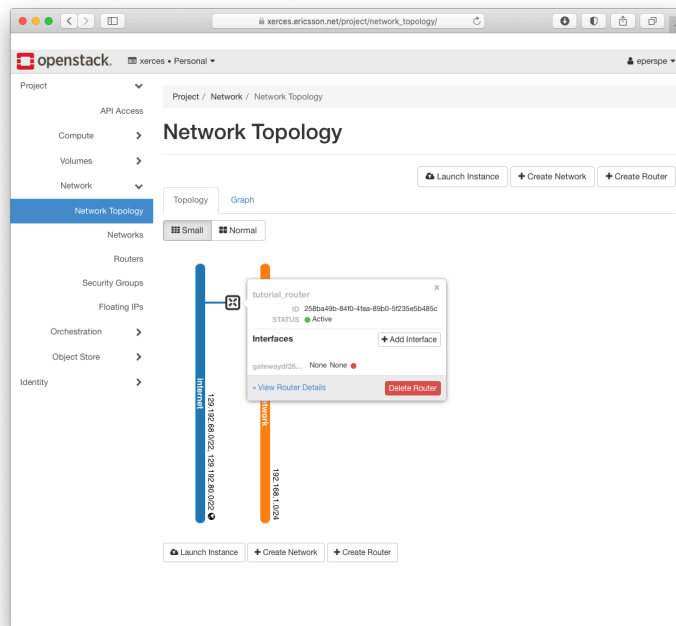
Add a router

- Select Network -> Network Topology. Now you should see "internet" and the new "tutorial_network"
- To make traffic flow between them you should add a router using the "Create Router" button
 - Enter **tutorial_router** as the name, select **internet** as the External Network, and click "Create Router"



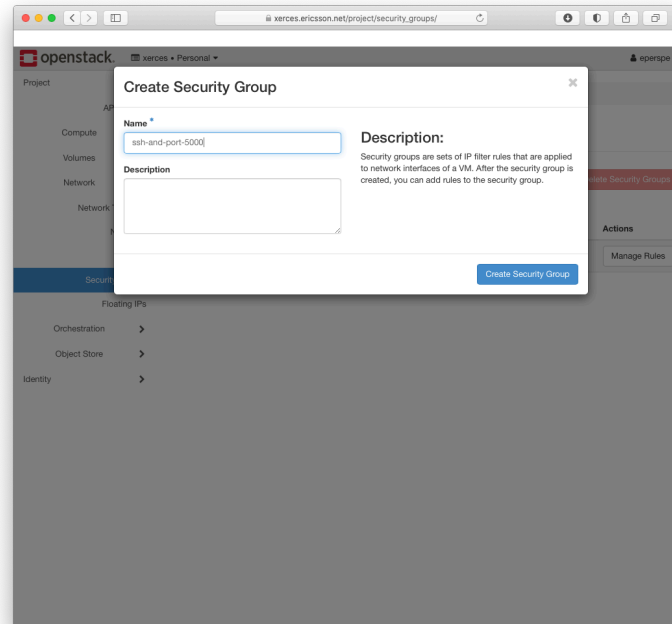
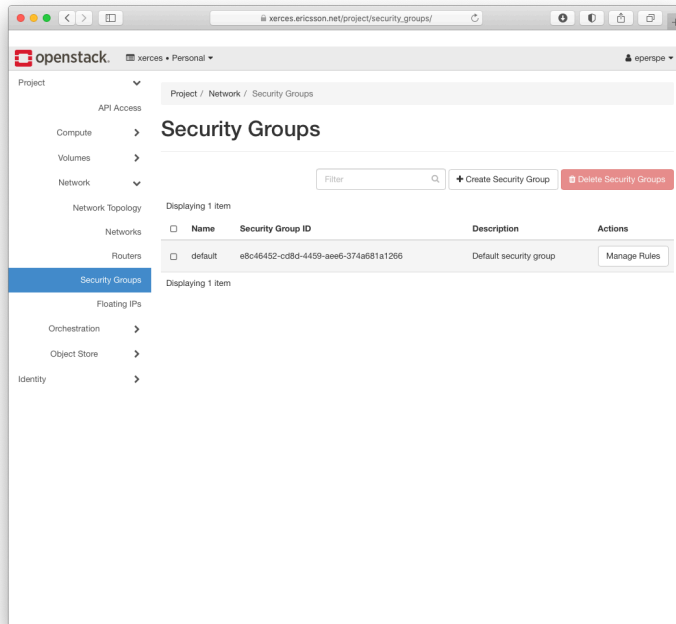
Connect router to subnet

- Select Network -> Network Topology.
- Hover over the router and click the "Add Interface button"
- Select the **tutorial_network: 192.168.1.0/24 (subnet)**, and click "Submit"
- You should now see the router connecting the two networks



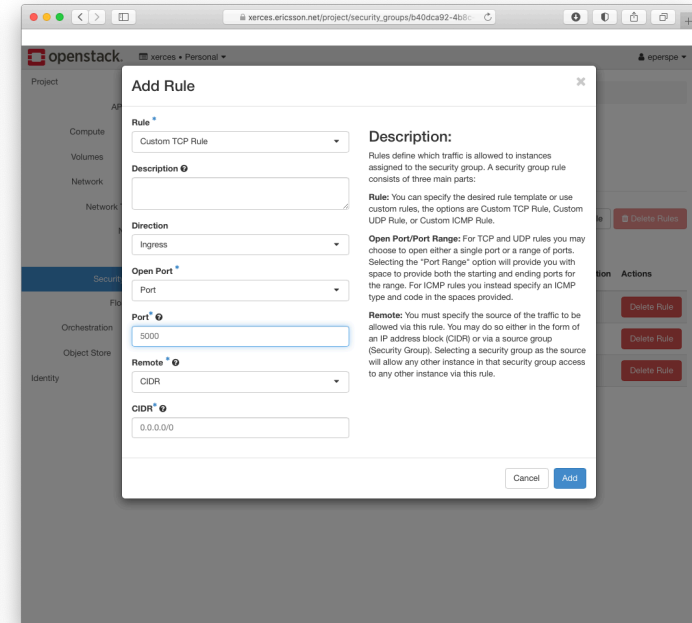
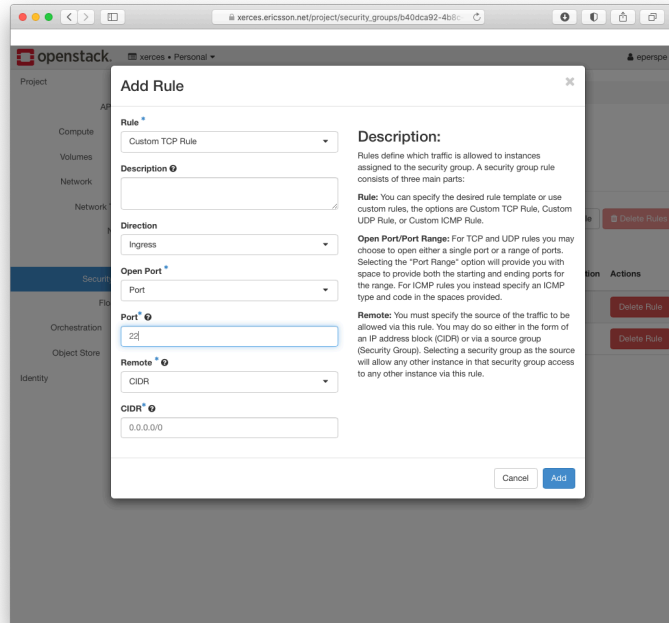
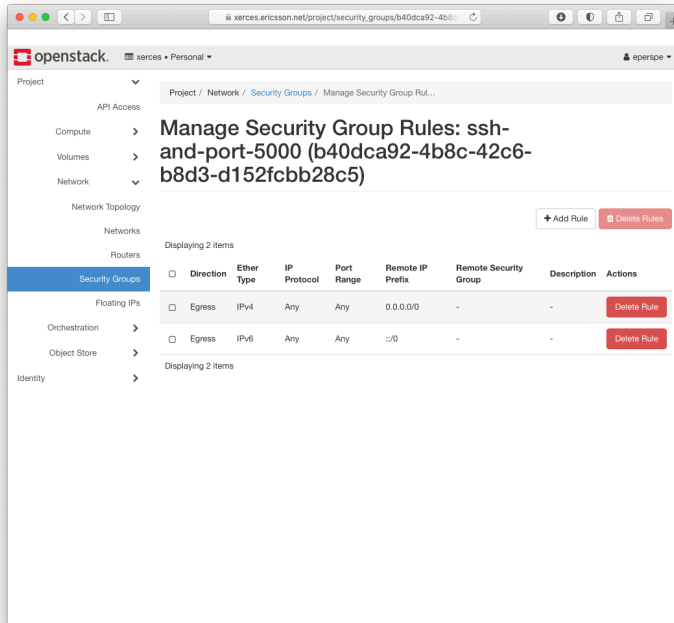
Configure security

- Everything is disabled so you need to open port 22 for SSH (remote login) and port 5000 for HTTP
- Click the "Create Security Group" button
 - In the dialog enter **ssh-and-port-5000** as the name and click "Create Security Group"



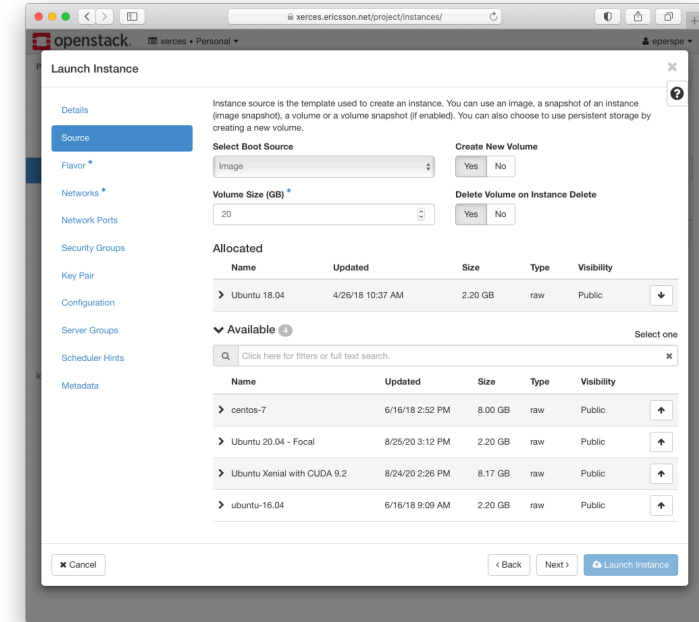
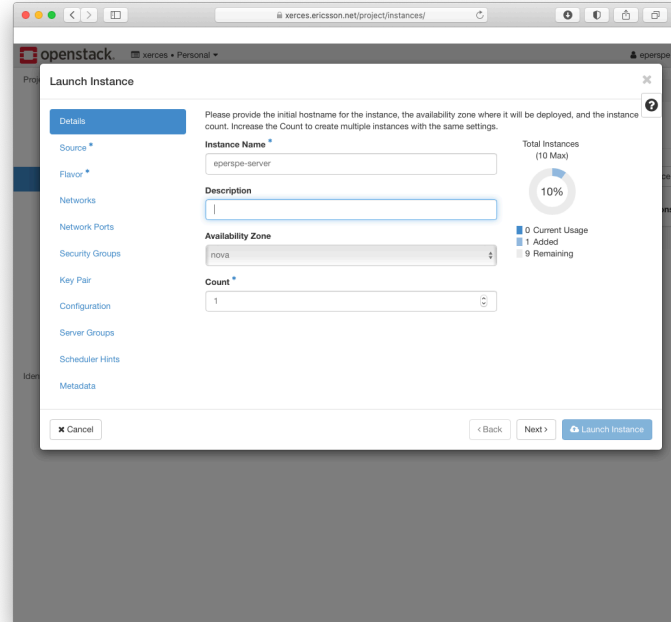
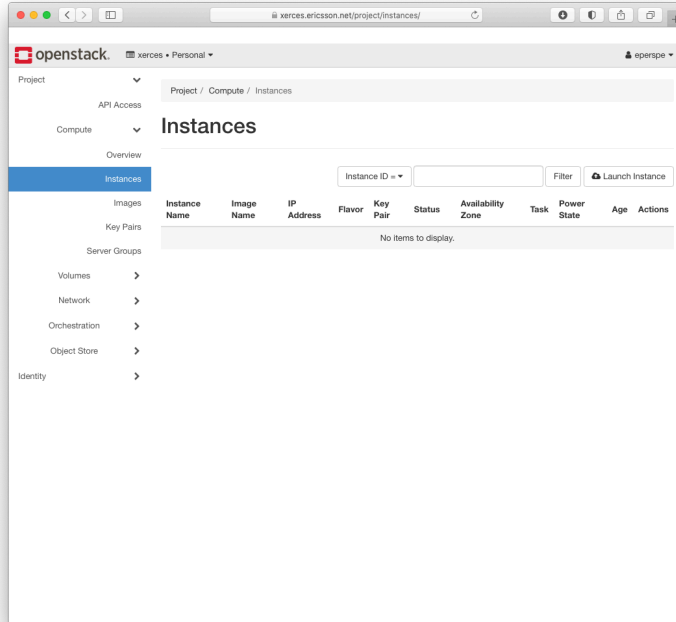
Add security rules

- With the new security group selected, click "Add Rule"
- In the dialog, enter **22** for the port (leave the rest as is) and click "Add"
- Repeat the process and add another rule
 - In the dialog, enter **5000** for the port (leave the rest as is) and click "Add"
- (In case you wonder, we'll run HTTP over port 5000 rather than 80)



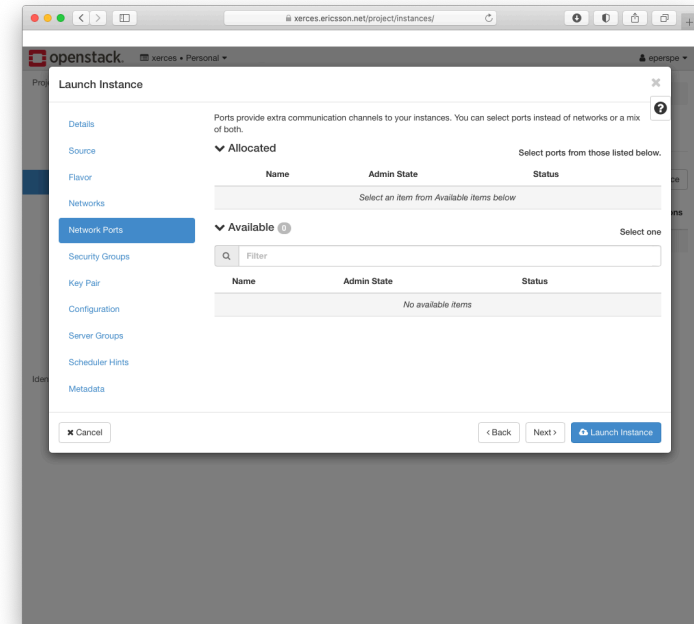
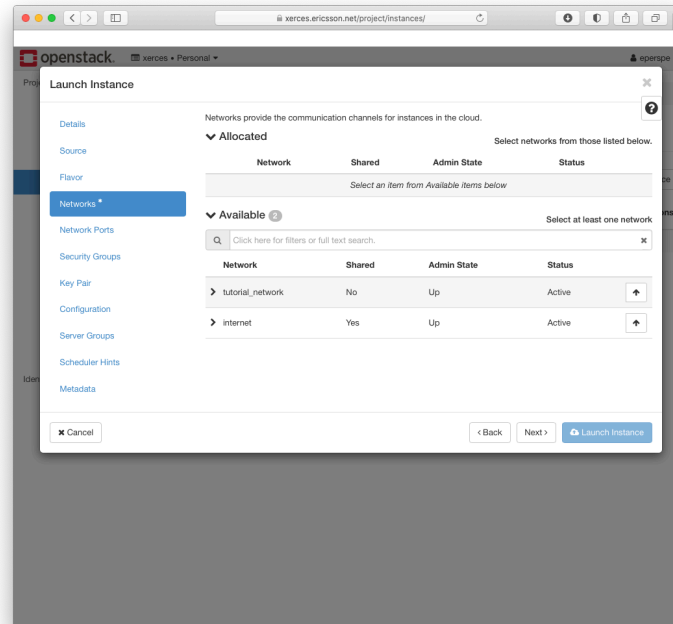
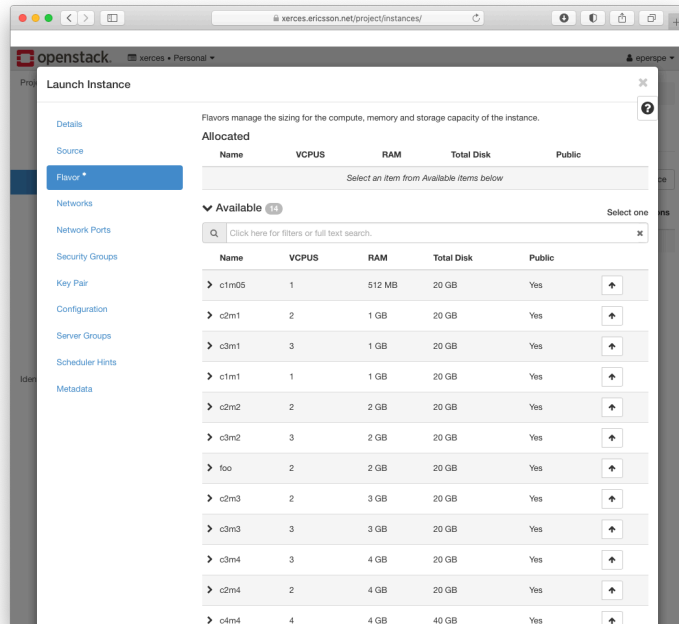
Add a VM (step 1 of 3)

- Select Compute -> Instances and click "Launch Instance"
- In the dialog Details tab enter **test-server** as name, and click "Next"
- In the Source tab
 - select **Image** as boot source, set volume size to **20**, and "Delete Volume on Instance Delete" to **Yes**
 - finally, **click the up-arrow** for "Ubuntu 24.04", and click "Next"



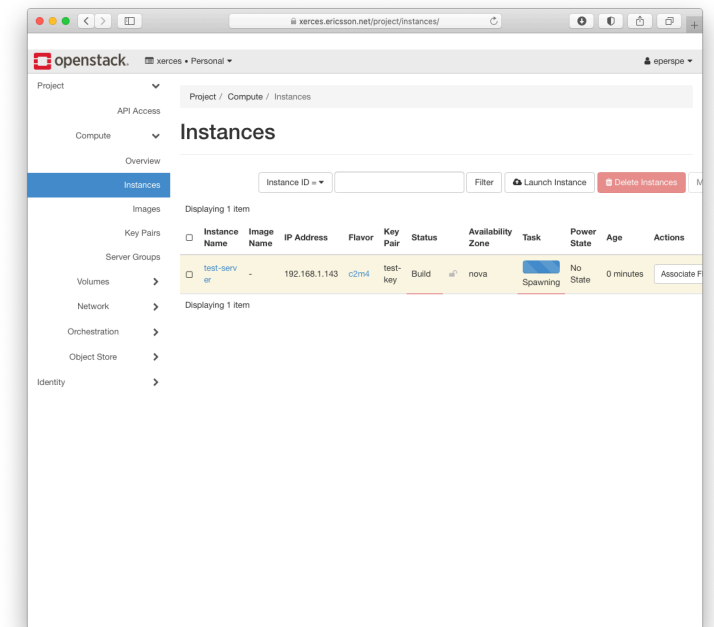
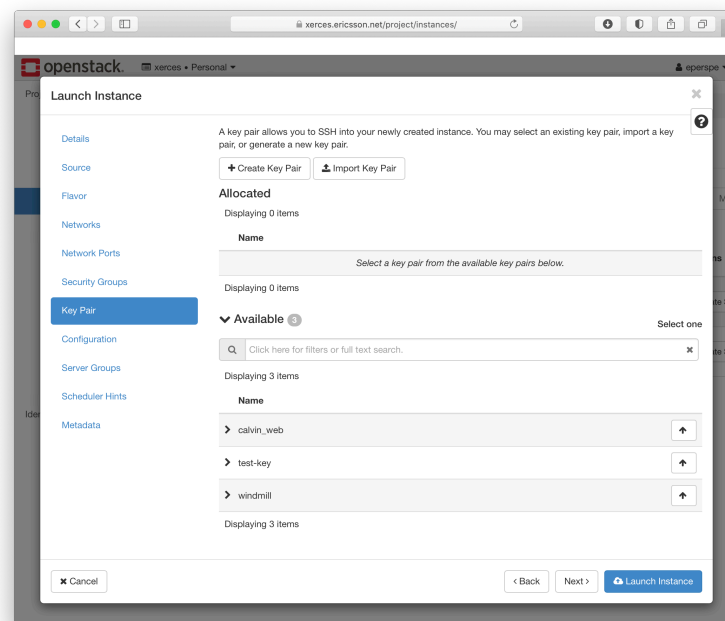
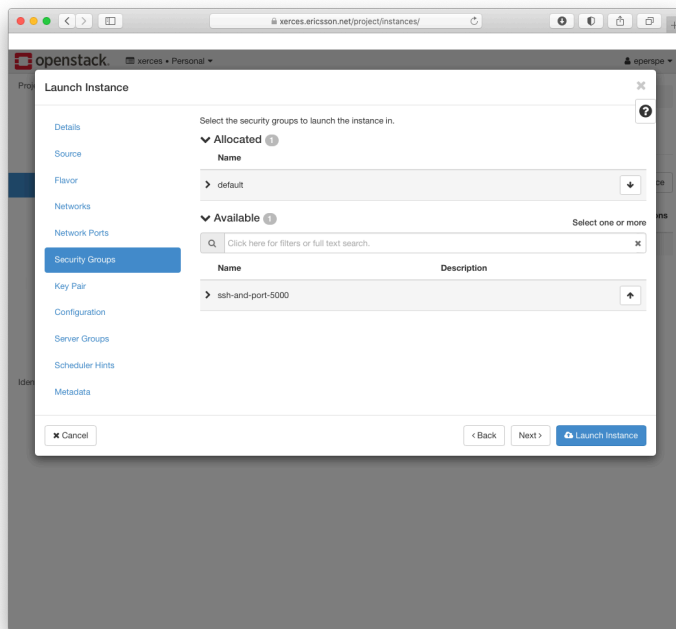
Add a VM (step 2 of 3)

- In the Flavour tab **click the up-arrow** for "c2m4" (i.e. 2 CPUs, 4GB RAM), and click "Next"
- In the Networks tab **click the up-arrow** for "tutorial_network", and click "Next"
- In the Network Ports tab there is nothing to do so just click "Next"



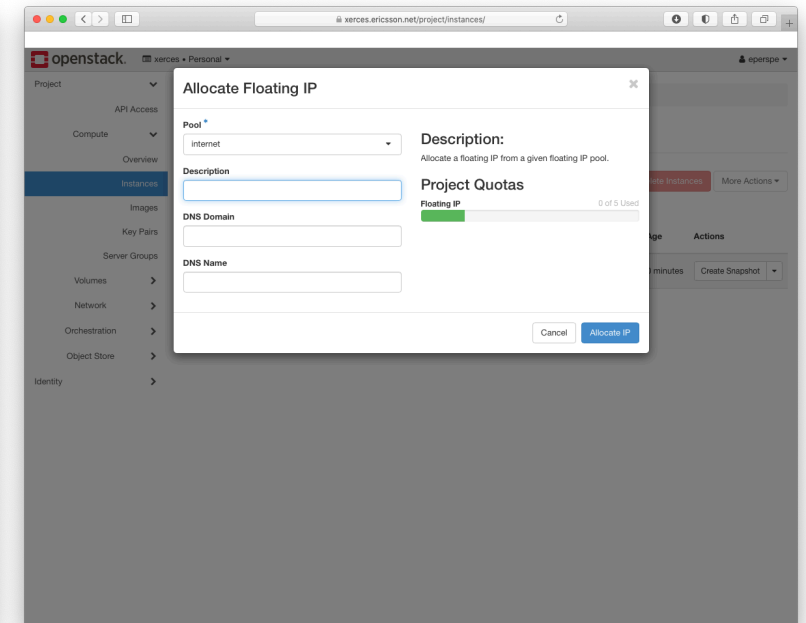
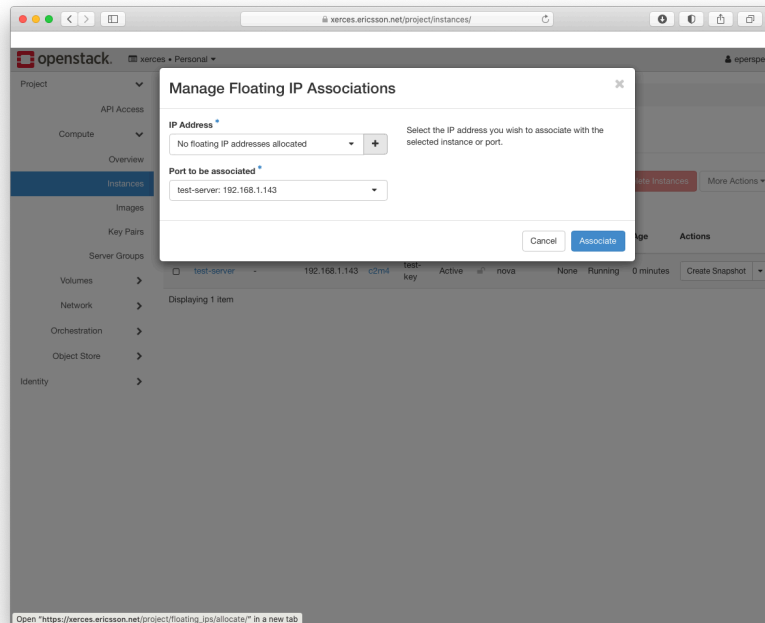
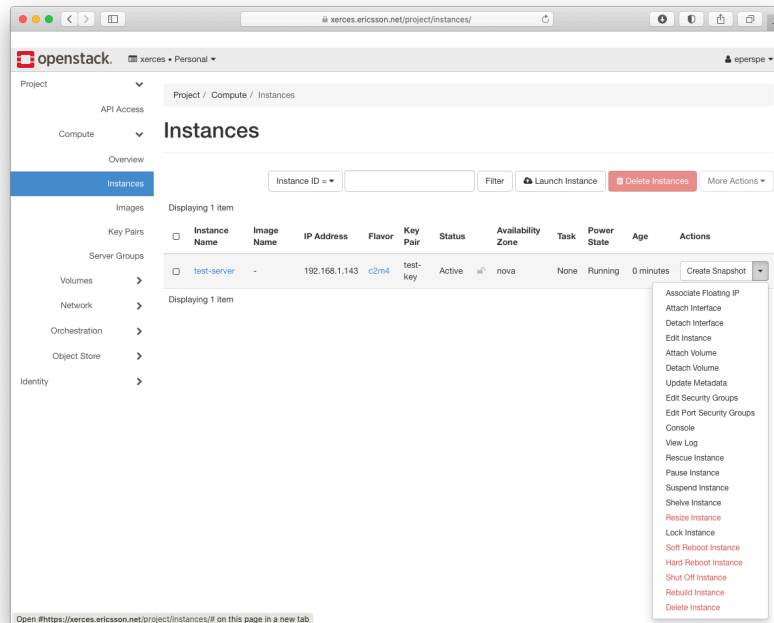
Add a VM (step 3/3)

- In the Security Groups tab **click the up-arrow** for "ssh-and-port-5000", and click "Next"
- In the Key Pair tab **click the up-arrow** for "test-key" (that you created earlier)
- The remaining tabs can be skipped, so click "Launch Instance"
- Select Compute -> Instances to see the VM being spawned



Associate an external address

- In Compute -> Instances select **Associate Floating IP** from the drop down menu at the far right
- In the dialog, select **test-server** for "Port to be associated", then press the **+** sign in "IP Address"
- In the second dialog, select **internet** for the "Pool" and press "Allocate IP"
- Back in the first dialog, now with an externally visible IP number, press "Associate"



Adding a Swift Object Store

- You need a shared object store (Swift) so let's create one
- In Object Store -> Containers click **+Container**
 - In the dialog enter your username as the name (it must be unique so your username is a good choice)
 - Press "Submit"

